INDUS BRAIN AI

AI-Powered Industrial Knowledge Intelligence Platform
ET AI Hackathon 2026

Abstract

INDUS BRAIN AI centralizes industrial documents such as maintenance reports, SOPs, manuals, inspection records and engineering drawings into a searchable AI-powered knowledge platform. Using Retrieval-Augmented Generation (RAG), OCR and vector search, engineers can ask natural-language questions and receive accurate answers with document references.

Problem Statement

Industries struggle with fragmented information spread across multiple systems, leading to slow document retrieval, knowledge loss, maintenance delays and compliance challenges.

Objectives

• Centralize industrial knowledge.
• Reduce search time.
• Improve maintenance planning.
• Preserve expert knowledge.
• Improve compliance.

Proposed Solution

Features include AI document search, PDF upload, OCR, AI chatbot, maintenance recommendations, compliance checker and analytics dashboard.

Technology Stack

Frontend: React + Tailwind CSS
Backend: FastAPI
AI: OpenAI GPT, LangChain
Vector DB: ChromaDB
OCR: Tesseract
Database: PostgreSQL

System Architecture

User → Web Dashboard → FastAPI Backend → RAG Engine → ChromaDB/OCR/GPT → Industrial Documents → AI Response

Workflow

1. Upload PDFs
2. OCR extracts text
3. Generate embeddings
4. Store in vector database
5. User asks questions
6. RAG retrieves relevant content
7. AI generates cited answers

Business Impact

Reduces downtime, speeds document retrieval, improves compliance, preserves organizational knowledge and supports better decision-making.

Future Scope

Mobile app, voice assistant, IoT integration, predictive maintenance and digital twin support.

Conclusion

INDUS BRAIN AI is a scalable AI-powered knowledge intelligence platform for industrial organizations.